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Meta-path based graph contrastive learning for micro-video recommendation

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ABSTRACT

Nowadays, micro-video sharing platforms have become popular tools for people creating and viewing micro-videos in daily life. The micro-video recommendation task has attracted significant attention from researchers, recently. The key to a high-quality micro-video recommendation system is learning meaningful user and micro-video representations. Recently, many graph neural networks (GNNs) are proposed to learn node representations in graphs, which could be useful for recommendation tasks. However, we argue that directly utilizing existing GNNs in micro-video recommendations is ill-posed. The reasons are: (1) most previous GNNs fail to capture the heterogeneity of heterogeneous graphs since they are designed for homogeneous graphs; (2) they overemphasize node proximity and may hurt the robustness of node embeddings. In contrast to previous work, we propose a meta-path-based graph contrastive learning network (MPGCL) that learns more meaningful user and video embeddings for recommendation. Specifically, we yield homogeneous graphs for user type and video type to better capture heterogeneity based on a well-designed meta-path-based random walk strategy. Furthermore, to learn more robust node embeddings that are less sensitive to noise, we propose a graph contrastive learning network on different types of homogeneous graphs, which maximizes the consistency between graph representations with different views. We do extensive experiments on three real-world datasets and the results show that our model can learn more meaningful embeddings for users and micro-videos and outperforms the strong baselines.

1. Introduction

The gradual popularity of micro-video sharing and creating platforms (e.g., Tiktok and Kwai) brings a large number of micro-videos distributed in online sources, which makes it difficult for users to immediately find micro-videos they are interested in among massive new micro-videos. In order to alleviate the problem of information overload and enhance user experience, designing high-quality micro-video recommendation systems has attracted significant attention from researchers recently (Chen et al., 2021; Liu et al., 2019; Wei et al., 2019).

The three main categories of currently used micro-video recommendation techniques are as follows: (1) Collaborative Filtering (CF) (Davidson et al., 2010; He et al., 2017; Huang et al., 2016; Wang et al., 2022) is a commonly used method that learns user and video representations from the user-video explicit (e.g., rating) and implicit (e.g., click) interaction history and recommends videos according to the similarity of users' behavior. However, these methods always suffer from data sparsity and cold start problems. (2) In order to tackle the problem of collaborative filtering, several methods are proposed to extract features of videos based on the contents (Cui et al., 2014; Deldjoo et al., 2016; Zhu et al., 2021), such as description texts and tags. (3) Deep neural network-based recommendation methods employ pre-trained deep learning models to represent contents into low-dimensional vectors to learn useful user and micro-video feature embeddings and gain some improvements for video recommendation (Chen et al., 2018; Guo et al., 2017). However, these methods focus on traditional online videos (e.g., YouTube, MSN Video) and heavily depend on extracting feature interactions based on the attribute information.

Recent advancements in graph neural networks (GNN) have shown that adopting GNNs can help enhance representation learning by capturing high-order interactions among nodes in a graph. These GNN-based methods aim to learn a node embedding by aggregating its neighbor features, such as GCN (Kipf & Welling, 2017), GraphSAGE (Hamilton et al., 2017), MMGCN (Wei et al., 2019), and LightGCN (He

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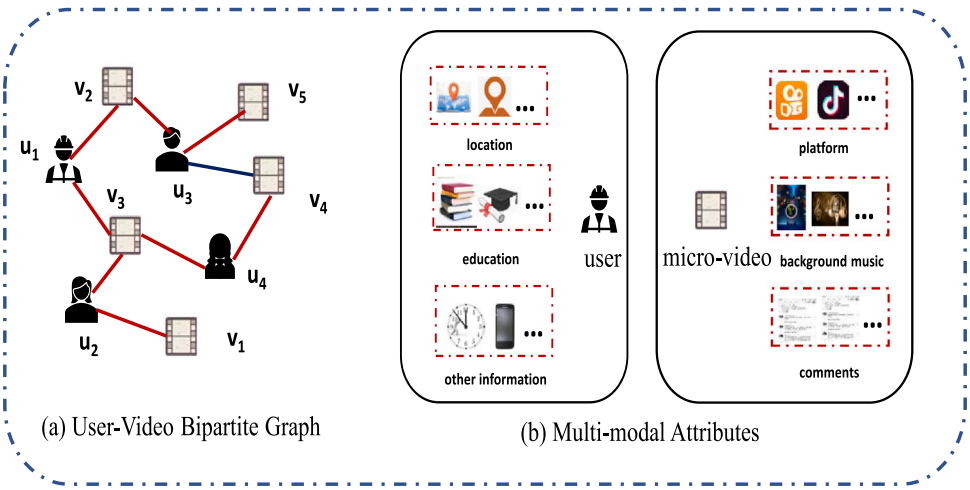


Fig. 1. Illustration of a heterogeneous user-video bipartite graph.

et al., 2020). Despite the fact that these GNN-based recommendation techniques have achieved great success, most of them focus on homogeneous graphs and fail to model the heterogeneity of graphs (e.g., user-video bipartite graphs) well. Actually, the heterogeneity of graphs is an intrinsic property in micro-video recommendation scenarios. Specifically, (a) there are various types in micro-video recommendation scenarios, such as user and micro-video types. (b) Different types of nodes are associated with multi-modal attributes. For example, in Fig. 1, the user's attributes contain locations, education, phone, and so on, while its neighbor video is associated with background music and comments. Intuitively, modeling the heterogeneity of users and micro-videos for embeddings may be significant for enhancing the performance of recommendations. Recently, heterogeneous graph neural networks are proposed to take the heterogeneous characteristics of graphs into consideration. However, these methods have some disadvantages. For example, meta-path-based methods, such as HAN (Wang et al., 2019) and HERec (Shi et al., 2019), generate different types of neighbors for each node according to meta-paths in heterogeneous graphs, which heavily depend on complicated meta-paths designed by experts. Another research line, including HetGNN (Zhang et al., 2019) and HHFAN (Cai et al., 2022), sample neighbors by a random walk with restart strategy. Although achieving promising performance on recommendation tasks, these methods rely heavily on the performance of random walk sampling. Therefore, how to model the heterogeneity of user-video graphs more reasonably is challenging.

Conventional homogeneous (e.g., GCN Kipf & Welling, 2017 and GraphSAGE Hamilton et al., 2017) and heterogeneous GNN-based methods (e.g., HAN (Wang et al., 2019), HERec (Shi et al., 2019) HetGNN (Zhang et al., 2019), HHFAN (Cai et al., 2022)) over-emphasize node proximity information. The underlying idea of these approaches is that nodes that have similar neighborhood sets should have similar latent embeddings. However, these methods are sensitive to noise information (e.g., noise edges) which could hurt the robustness of node embeddings. In fact, noise edges may be caused by users clicking on micro-videos that they are not interested in for some reason (e.g., micro-videos recommended by friends). For example, in Fig. 1, there is a noise edge between u_3 and v_4. Based on traditional GNN-based recommendation methods, u_3 and u_4 would be mapped to similar representations due to first-order node proximity (e.g., u_3, v_4, u_4). In other words, there should be some common preference between them. Hence, these methods may recommend the wrong micro-videos (e.g., v_5) for u_4 according to u_3. More recently, emerging graph contrastive learning (GCL) techniques take advantage of GNNs and contrastive learning to effectively improve the robustness of graph representation learning. For example, GraphCL (You et al., 2020) is a novel method for learning robust representations on the graph which has obtained great improvements in the tasks of node and graph classification. This method aims to learn a graph encoder that maximizes the consistency between representations encoded from different views. However, most graph contrastive learning models cannot be used for micro-video recommendation tasks directly since they are applied to homogeneous graphs. Therefore, how to learn robust representations which are less sensitive to noise for nodes in graphs is a challenging problem.

To address the above issues, in this paper, we propose a novel Meta-path Based Graph Contrastive Learning Network for Micro-video Recommendation (MPGCL) which can effectively capture the heterogeneity and learn more robust representations on user-video bipartite graphs. Specifically, we learn comprehensive node embeddings by the following step: (1) To effectively model user-video bipartite graphs without losing the heterogeneity information, we employ a specially designed meta-path-based random walk strategy to construct homogeneous graphs for user type and micro-video type and utilize a GCN model on each type of homogeneous graph for representation learning. We just introduce simple meta-paths to build homogeneous graphs to model heterogeneity, which is different from meta-path-based methods that design complicated meta-paths to explore heterogeneous neighbors for information aggregation. (2) To learn more robust node embeddings that are less sensitive to noise, we introduce a contrastive learning method for each type of homogeneous graph, which maximizes the consistency between node representations with different views. (3) We train a basic GCN on the user-video bipartite graphs by a designed constraint loss and train the type-specific GCN by the unsupervised contrastive learning loss. We design the final loss as the combination of the above-mentioned losses and a well-designed supervised loss, such that our model can be trained in an end-to-end fashion.

In conclusion, the key contributions of our method are summarized as follows:

- We propose a novel micro-video recommendation model, namely Meta-path Based Graph Contrastive Learning Network (MPGCL). To effectively capture the heterogeneity, our model learns node representations on user and video homogeneous graphs constructed by a well-designed random walk strategy based on meta-path.
- We introduce a specially designed contrastive learning method for representation learning, which could learn robust node embeddings by maximizing the consistency between different views of graph representations.
- Our method is evaluated on three real-world datasets including Kwai, Tiktok, and MovieLens. Experiments on these datasets show that our method yields great improvements compared with strong baselines.

2. Related work

2.1. Video recommendation

Recently, researchers have shown great interest in video recommendation since the gradual development of video-sharing platforms. Traditional video recommendation methods mainly contain three categories: collaborative filtering approaches (Davidson et al., 2010; He et al., 2017; Huang et al., 2016; Wang et al., 2022), content-based approaches (Cui et al., 2014; Deldjoo et al., 2016; Zhu et al., 2021), and feature combination-based approaches (Chen et al., 2018; Guo et al., 2017). Collaborative Filtering (CF) is a typical approach that learns embeddings for users and videos based on the user-video interaction history. For example, Huang et al. (2016) develop an online CF method that considers the implicit feedback of users for video recommendation. Unfortunately, these methods suffer from data sparsity and cold start problems. To deal with data sparsity, several methods analyze user and item content for feature extraction. Cui et al. (2014) integrate text information of videos and social interactions of users into video and user representations, respectively. Zhu et al. (2021) introduce a semi-autoencoder to preserve items' attribute information into embeddings to alleviate the sparsity problem. Recently, many feature combination-based approaches are proposed to incorporate factorization-machine to learn feature interactions of users and video for better video recommendation. DeepFM (Guo et al., 2017) introduces deep learning to the traditional factorization machines in a unified framework, which can model both high and low feature interactions. However, most of the video recommendation methods focus on traditional online videos (e.g., YouTube, MSN Video).

2.2. Graph neural network based recommendation

Due to the powerful representation learning ability, graph neural network (GNN) has shown its effectiveness in many recommendation applications in different fields, such as video recommendation (Cai et al., 2022), news recommendation (Hu et al., 2020), product recommendation (Wang et al., 2020), music recommendation (Hu et al., 2021), and so on. GNN-based recommendation methods first model users and items into a bipartite graph where the edges indicate the direct relationships between nodes. Then, they design an appropriate propagation function (e.g., convolution) to propagate node features to neighbor nodes and update user and item embeddings by an aggregate function. Homogeneous graph neural networks-based methods, such as GCN (Kipf & Welling, 2017), GraphSAGE (Hamilton et al., 2017), GAT (Velickovic et al., 2018), aggregate neighbor information for each node by convolutional operation, which may improve the performance of recommendation. For example, PinSage (Ying et al., 2018) designs a graph convolution method that incorporates an efficient random walk strategy to generate embeddings of nodes. This method can not only model graph structure but also consider node feature information. MMGCN (Wei et al., 2019) designs a graph convolution network that learns user and micro-video embedding from multi-modal features. However, this method treats all neighbors equally when aggregating information. LightGCN (He et al., 2020) introduces a graph network that utilizes an efficient and simple graph convolution operation where feature transformation and nonlinear function of standard GCN are removed. UltraGCN (Mao et al., 2021) takes a further step by skipping message passing of multiple layers for the efficient recommendation. Although effective, most of these methods focus on homogeneous graphs and fail to model the heterogeneity of graphs. More recently, heterogeneous graph neural networks take the heterogeneous characteristics of graphs into consideration (e.g., HAN, MAGNN, HERec, MAF-GNN Song et al., 2022, HHFAN Cai et al., 2022, and K-NCR Sang et al., 2021). Meta-path-based methods, such as HAN (Wang et al., 2019), MAGNN (Fu et al., 2020), and HERec (Shi et al., 2019), learn a meaningful representation for each node by aggregating different types of neighbor information generated based on well-designed meta-paths in heterogeneous graphs. For example, HAN (Wang et al., 2019) is designed for heterogeneous graphs which uses a graph neural network with hierarchical attention to evaluate weights of nodes and meta-paths for node representations. However, these meta-path-based methods heavily depend on complicated meta-paths designed by experts. Another research line, including HetGNN (Zhang et al., 2019) and HHFAN (Cai et al., 2022), samples a fixed size of heterogeneous neighbor sequence for each node by a random walk with restart strategy and groups each type of node for representation learning. For example, HHFAN (Cai et al., 2022) is a state-of-the-art approach that learns node representation from multi-views by utilizing hierarchical feature aggregation on the heterogeneous user-video graph for the micro-video recommendation. Although achieving promising performance on recommendation tasks, these methods rely heavily on the performance of random walk sampling. To solve these problems, our MPGCL constructs homogeneous graphs for different types of nodes by utilizing simple meta-paths, which can effectively capture the heterogeneity of user-video bipartite graphs.

2.3. Contrastive learning

Contrastive learning (CL) is currently a promising method for representation learning. Existing methods including DIM (Hjelm et al., 2019), AMDIM (Bachman et al., 2019), and SimCLR (Chen et al., 2020) are unsupervised learning methods that maximize consistency between inputs and their augmentations. Recent methods have attempted to transfer CL to a graph which can help obtain high-quality representations of nodes. However, designing different local representations of a graph is non-trivial. DGI (Velickovic et al., 2019) proposes to contract the graph representation with node representations by maximizing mutual information. InfoGraph (Sun et al., 2020) follows the same philosophy as DGI. The main differences between them are the navigate sample strategy and the encoder choice. Cao et al. (2021) proposes an attention mechanism for local representations. Hassani and Ahmadi (2020) contract node and graph representation from a multi-view perspective. You et al. (2020) propose GraphCL, which aims to learn robust representations for graphs by maximizing feature consistency between different augmented views. Although these methods have achieved great success, most graph contrastive learning models cannot be used for micro-video recommendation tasks directly since they are applied to homogeneous graphs. In our work, we leverage the graph contrastive learning mechanism for each type of homogeneous graph, which maximizes the consistency between nodes representations with different views to learn more robust user and video representations for micro-video representations.

3. Proposed method

3.1. Problem definition

In the micro-video recommendation scenario, there are user set and video set which we denote as $U = \{u_1, u_2, \dots, u_{|U|}\}$ and $V = \{v_1, v_2, \dots, v_{|V|}\}$, respectively, where $|U|$ is the number of users and $|V|$ is the number of the videos. Each $u \in U$ is a user which consists of a set of attributes $\{a_u^1, a_u^2, \dots, a_u^l\}$ to describe it, such as age, name, gender, and so on. Similarly, there are some meta-data associated with a video $v \in V$ which is denoted as $\{a_v^1, a_v^2, \dots, a_v^m\}$. The meta-data contains topics of micro-video, content description, title of micro-video, tags of micro-video, and so on. Let $I \in \mathbb{R}^{|U|\times|V|}$ denote the interaction matrix according to users' implicit feedback with video (click a video), where entry $i_{ij} = 1$ if user u_i shows preference for video j, $i_{ij} = 0$ otherwise.

To preserve interaction information between users and videos, we build a user-video bipartite graph $\mathcal{G} = \{U, V, S\}$ of user and video, where U, V are user set and video set. Let S denote the adjacency of

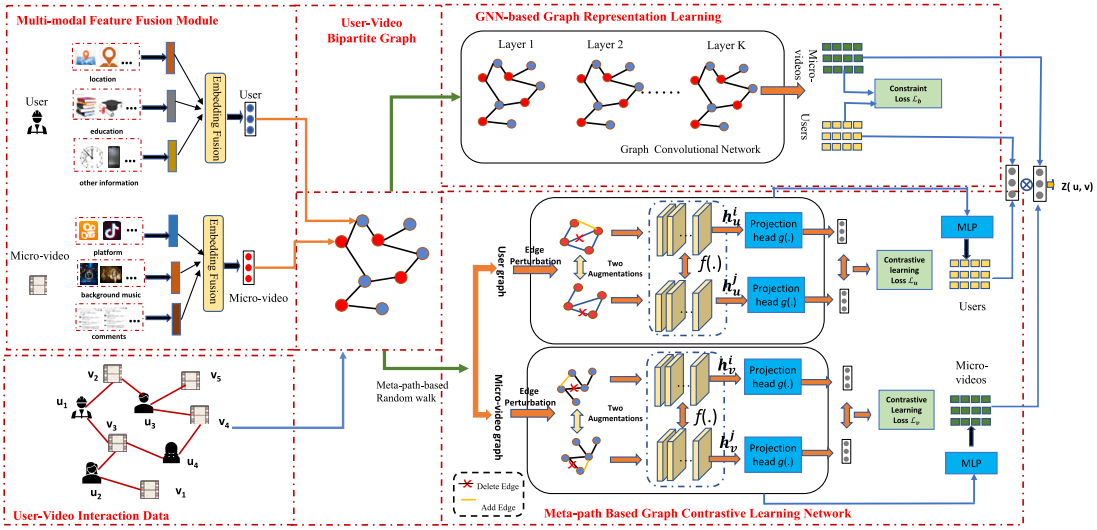


Fig. 2. Overall framework of the MPGCL algorithm.

the bipartite graph. Intuitively $S = I$, where $S_{ij} = 1$ represents node u_i has relation to node v_j, $S_{ij} = 0$ otherwise.

Given a user-video bipartite graph $\mathcal{G}$, feature matrices $X_u \in \mathbb{R}^{|U|\times d_1}$ and $X_v \in \mathbb{R}^{|V|\times d_2}$ of users and videos, the goal of our micro-video recommendation is to learn a function f for predicting how much preference a user to a video according to the learned embeddings of users and videos. f is denoted as: $S = f(\mathcal{G}, X_u, X_v)$, where $S \in \mathbb{R}^{|U|\times|V|}$ is the predicted preferences of users to videos.

3.2. Overall framework

We give the overall framework for our proposed model Meta-path Based graph contrastive learning network for micro-video recommendation (MPGCL) in this section. Fig. 2 demonstrates the architecture of MPGCL, which includes the following key parts.

- **Multi-modal Feature Fusion:** In order to better extract interactions of users and videos, we build a bipartite graph of user nodes and video nodes. Instead of randomly initializing the embeddings of nodes in the bipartite graph, we first utilize an embedding fusion layer to initialize them by fusing their corresponding multi-modal attribute features.
- **GNN-based Graph Representation Learning Network:** Based on the constructed user-video bipartite graph, a GNN-based graph representation encoder is designed to learn embeddings for users and videos based on their interactions. The network first takes initial node embeddings as inputs and employs a well-designed GNN-based encoder for interaction extraction. This part is demonstrated at the top of Fig. 2.
- **Meta-path Based Graph Contrastive Learning Network:** This part first yields type-specific homogeneous graphs for user type and micro-video type by utilizing a specially designed meta-path based random walk strategy. Next, a homogeneous graph convolution network is used to propagate information on type-specific homogeneous graphs to avoid losing heterogeneity. To be able to obtain more robust node embeddings that are less sensitive to noise, we design a type-specific graph contrastive learning network that maximizes the consistency between graph representations with different views. The designed contrastive loss function is optimized in an unsupervised manner over all nodes.
- **Model Optimization and Prediction:** We fuse the results of the GNN-based graph representation learning network and the meta-path-based graph contrastive learning network for the final embeddings of users and micro-videos. Then, the parameters of the model are optimized by jointly learning GCN constraint loss, contrastive loss, and Bayesian Personalized Ranking (BPR) loss. Finally, based on the learned representations, a prediction layer is designed for predicting the preference of a user to a video.

3.3. Multi-modal feature fusion

Instead of randomly initializing user and video embeddings, we first utilize an embedding fusion layer to initialize them by fusing their corresponding multi-modal attribute features, respectively. To be more specific, the attribute features are pre-trained by different methods according to their types. For example, attributes with image type are pre-trained by CNN and the textual attributes are pre-trained by BERT. Therefore, each attribute of user or video is mapped to a low-dimensional vector $\mathbf{a}_u$ or $\mathbf{a}_v$. Then, similar to many recommendation systems, we get user or video embedding by concatenating the attribute features. For example, we concatenate a video's attributes as:

$$\mathbf{f}_i = concat(\mathbf{a}_v^1, \mathbf{a}_v^2, \ldots, \mathbf{a}_v^l) \tag{1}$$

where $\mathbf{f_i}$ represents the ith row of the video embedding A_v. Such operation can also be replaced by many typical neural networks, such as CNN and LSTM. The fused embeddings for micro-video nodes are generated by a fully connected neural network:

$$X_v = W_v A_v + b \tag{2}$$

The fusion process of user nodes X_u is similar with X_v.

3.4. GNN-based graph representation learning

Learning user and video latent representation is the core of our recommendation system. In order to better extract user and video interaction, we introduce a graph learning method following the idea of GCN to learn the node representations of the user-video bipartite graph. The proposed method combines LightGCN (He et al., 2020) and UltraGCN (Mao et al., 2021) for representation learning, which is a basic encoder in our model. We first follow LightGCN to aggregate features for nodes, which could be directly used on user-video heterogeneous graphs without feature transformation and nonlinear activation. Taking the micro-video node as an example, we perform graph convolution operation iteratively as follows:

$$x_v^{k+1} = \frac{1}{d_v + 1} x_v^k + \sum_{u\in\mathcal{N}(v)} \frac{1}{\sqrt{|d_v + 1|}\sqrt{|d_u + 1|}} \tag{3}$$

where $\mathcal{N}(v)$ is the neighbor nodes of a video v, d_v and d_u are the degree of a video v and neighbor node u, respectively. $\sqrt{|d_v+1|}\sqrt{|d_u+1|}$ is the symmetric normalization term which is the same as standard GCN (Kipf & Welling, 2017). To learn our GNN-based encoder, inspired by UltraGCN, we derive the loss function as:

$$\mathcal{L}_b = -\sum_{v\in V}\sum_{u\in\mathcal{N}(v)} \beta_{u,v}\log(\sigma(e_u^\top e_v)), \quad \beta_{u,v} = \frac{1}{d_v}\sqrt{\frac{d_v+1}{d_u+1}} \tag{4}$$

where σ is s the sigmoid function.

3.5. Meta-path based graph contrastive learning network

The previous step of the GNN-based encoder aims to extract semantic interactions of nodes for representation learning. However, as mentioned above, Traditional GNN-based methods over-emphasize node proximity information which are sensitive to noise. To learn more robust node representations, we propose a meta-path-based graph contrastive learning network for node representation. To get a more meaningful representation for each node, we first construct homogeneous graphs for user type and micro-video type. Then, we perform contrastive learning on different types of homogeneous graphs. Specifically, our model first describes how to generate different views of user and video homogeneous graphs. Then, we learn node representations for each augmented graph by a GNN-based graph encoder. Finally, to obtain robust representations, we utilize a normalized temperature-scaled cross-entropy loss (NT-Xent) (You et al., 2020) on different types of homogeneous graphs to maximize the consistency between different graph augmentations with different structural views.

3.5.1. Homogeneous graph construction

Existing graph neural network methods, such as GCN (Kipf & Welling, 2017), GraphSage (Hamilton et al., 2017), and GAT (Velickovic et al., 2018), focus on homogeneous graphs where the nodes are of the same type. These methods cannot be directly utilized in recommendation systems since nodes in the user-video bipartite graph have different types. Thus, if a node representation updates its feature by aggregating information from its direct neighbors in the user-video graph indiscriminately, it may lose type-specific information due to the negligence of node type. To solve this issue, we present a random walk strategy based on the meta-path to transform a bipartite graph into a type-specific homogeneous graph.

We take nodes whose type is video for example. Our goal is to generate a meaningful homogeneous graph for videos that can represent the semantic relation of the original graph. Specifically, we first define meta-path schemes related to video to generate meta-path-based sequences. Then, inspired by Shi et al. (2019), we utilize a meta-path-based random walk to generate node sequences based on meta-path schemes. Finally, we delete nodes that have different types of videos. The remaining video nodes form a homogeneous graph. We detail each step in the following.

Given a user-video bipartite graph $\mathcal{G} = \{U, V, S\}$, a meta-path ρ can be defined as: $A^1 \xrightarrow{R_1} \dots A^t \xrightarrow{R_t} A^{t+1} \dots \xrightarrow{R_l} A^{l+1}$, where A^t and R_t denote the node and relation type. According to this scheme, we can derive some meta-paths where the type of the starting node is a video (V). (1) V–U–V refers to that two videos being watched by the same user. (2) U–V–U means that two users watch the same video.

$$P(n_{t+1}|n_t,\rho) = \begin{cases} \frac{1}{|\mathcal{N}_{A^{t+1}}(n_t)|}, & (n_t, n_{t+1}) \in \mathcal{E},\ \phi(n_{t+1}) = A^{t+1} \\ 0, & otherwise \end{cases} \tag{5}$$

where $\mathcal{N}_{A^{t+1}}$ is the one-hop neighbors for node n_t whose type is A^{t+1}. A random walk sampling operation will iteratively sample nodes following the pre-defined meta-paths until the sequence reaches a pre-defined length. Finally, we could generate multiple sequences according to the scheme of meta-paths.

Since we aim to construct a homogeneous graph for micro-videos, after generating sequences based on the above random walk strategy, different types of micro-video nodes will be deleted. The remaining random walk sequence could contain video nodes only. Thus, we can get a homogeneous graph for videos, which is denoted as $\mathcal{G} = \{V, S^{\mathrm{v}}\}$. A similar process is also adopted to generate a homogeneous graph for users, which is denoted as $\mathcal{G} = \{U, S^{\mathrm{u}}\}$.

3.5.2. Graph data augmentation

In order to make sure all nodes on each type-specific homogeneous graph obtain a robust representation that is less sensitive to noise, we first sample a sub-graph for each node and generate augmentations for the sampled sub-graph. To generate different views of sub-graph, we design an edge perturbation augmentation method with different ratios which means that the structure of the graph has certain robustness to the relational pattern perturbation.

Taking the video homogeneous graph as an example, we first define the perturbation matrix $M_{ij}^v = Bernoulli(\eta)$, where each entry means whether to perturb the corresponding entry of adjacency matrix S^v, η is the perturbation ratio. Then, we could obtain two correlated views $\hat{\mathcal{G}}_i$ and $\hat{\mathcal{G}}_j$ of the graph by exclusive operation with different ratios:

$$\hat{\mathcal{G}}(U, V, S') = S^v \oplus M^v \tag{6}$$

3.5.3. Type-specific node representation encoder

After obtaining two augmentations $\hat{\mathcal{G}}_i$ and $\hat{\mathcal{G}}_j$, we further learn high-order semantic representations for users and videos on the sub-graph. We adopt a commonly used graph encoder-GCN (Kipf & Welling, 2017) to learn the node embeddings in our work. As illustrated in Fig. 2, after building the type-specific homogeneous graph for users and micro-videos respectively, we feed each augmented homogeneous graph into a GCN model with multiple layers. Each node aggregates the features from its k-hop neighborhood nodes at the kth layer. Then, we get the representation of each node with high-order interaction information. Let X_v denote the feature matrices of videos. Formally, we define the kth layer propagation function as follows:

$$H_v^{(k)} = f(X_v, \hat{S}^v) = \sigma(\hat{D}^{-\frac{1}{2}} S^v \hat{D}^{-\frac{1}{2}} X_v^{(k)} W_v^{(k)}) \tag{7}$$

where $\hat{S}^v = S^v + I_n$ denotes the adjacency matrix with inserted self-loops, $\hat{D}^{-\frac{1}{2}} S^v \hat{D}^{-\frac{1}{2}}$ is the symmetric transition matrix and D is the diagonal matrix of node degrees. σ is a LeakyReLU nonlinear function. After stacking aggregation layers for K times, we get the final embedding for each node, and the higher-order neighborhood node features are aggregated to each node embedding. After the GNN encoder, we encode each node in the sub-graph into embedding h_v (video node).

3.5.4. Contrastive loss function

As mentioned above, traditional methods usually learn GNN-based encoder by predicting contextual node information based on its limited neighborhoods. The underlying assumption is that adjacency nodes are close in the representation space. However, it over-emphasizes proximity information. As a result, the learned node embeddings are sensitive to noise, which is of great importance for recommendation task. Hence, to address the limitations, we propose to adopt NT-Xent on each learned homogeneous graph to maximize the consistency between different graph augmentations with different structural views. By doing so, we could learn more robust node representations.

We randomly sample a minibatch of N nodes on the homogeneous graph. For each node, we have a positive pair of augmented sub-graph, resulting in $2N$ augmented sub-graph with their central nodes. we treat the other $2(N-1)$ central nodes of other augmented graphs within a minibatch as negative examples. With two different representations for each node on each homogeneous graph, we define our learning method

for the encoder as a NT-Xent loss following GraphCL (You et al., 2020):

$$\mathcal{L}_v = -\log \frac{\exp(D(h_v^{\hat{\mathcal{G}}_i}, h_v^{\hat{\mathcal{G}}_j})/\tau)}{\sum_{k=1}^{N} \exp(D(h_v^{\hat{\mathcal{G}}_i}, h_v^{\hat{\mathcal{G}}_k})\tau} \tag{8}$$

where $h_v^{\hat{\mathcal{G}}_i}$, and $h_v^{\hat{\mathcal{G}}_j}$ are the central nodes of two augmented graphs $\hat{\mathcal{G}}_i$ and $\hat{\mathcal{G}}_j$, τ is a temperature factor, D is a learnable score function which is denoted as follows:

$$D(h_v^{\hat{\mathcal{G}}_i}, h_v^{\hat{\mathcal{G}}_j}) = g(h_v^{\hat{\mathcal{G}}_i} W h_v^{\hat{\mathcal{G}}_j}) \tag{9}$$

where g is a non-linear projection head.

Similarly, we obtain the loss $\mathcal{L}_u$ for the user's homogeneous graph. Finally, we combine $\mathcal{L}_u$ and $\mathcal{L}_v$ into a single unsupervised loss:

$$\mathcal{L}_{un} = \mathcal{L}_u + \mathcal{L}_v \tag{10}$$

4. Model optimization

We fuse the results of the GNN-based graph representation learning network and the meta-path-based graph contrastive learning network for the final embedding of each node. We denote their representation as h_u and h_v. Since we aim to predict the likelihood that a user prefers a micro-video, we predict each user's rating to a candidate micro-video by calculating the inner product.

$$z(u, v) = \sigma(h_u^T h_v) \tag{11}$$

where $\sigma(x) = \frac{exp(x)}{1+exp(x)}$ is the sigmoid function, h_u and h_v are the learned representations for user and micro-video, $z(u, v)$ is the score to predict the preference of user u to micro-video v.

Given a dataset $\mathcal{D} = \langle u_i, v_i \rangle$, where u_i is a user and v_i is a micro-video. For any positive user micro-video pair in dataset $\mathcal{D}$, we sample a number of negative v_j according to the behavior of u_i (i.e no click or like). Then, we get a set of triples for training.

$$\mathcal{R} = \{(u, v, v') | (u, v) \in \mathcal{G}, (u, v') \notin \mathcal{G}\} \tag{12}$$

Following previous recommendation methods, we use Bayesian Personalized Ranking (BPR) (Rendle et al., 2009) to learn the GCN encoder in a supervised manner, which assumes that users prefer micro-videos that they have interactions than micro-videos that they do not have before. Therefore, BPR is formulated as follows:

$$\mathcal{L}_{su} = \sum_{(u,v,v') \in \mathcal{R}} -\ln(z(u, v) - z(u, v')) + \lambda \|\Theta\|_2^2 \tag{13}$$

where λ is the regularization weight to control the strength of L_2 to prevent overfitting, Θ is a parameter of the model.

Based on the above discussion, we defined the overall optimization function by combining three parts of losses and jointly train them. The total loss is formulated as:

$$\mathcal{L} = \alpha \mathcal{L}_b + \beta \mathcal{L}_{un} + \gamma \mathcal{L}_{su} \tag{14}$$

where α, β and γ are the hyper-parameters to assign the weight among these parts. The supervised loss is optimized over user-video pairs contained in dataset while the unsupervised losses are optimized over all nodes in the graph. Therefore, MPGCL will be optimized from the two aspects.

5. Experiments

5.1. Experimental setup

5.1.1. Datasets

Our model is evaluated on three real-world datasets including Kwai, Tiktok, and MovieLens. The statistics of these datasets are illustrated in Table 1. We detail all datasets as follows.

Table 1
Statistics of datasets.

Dataset	User	Micro-video	Interaction	Sparsity
Kwai	169,878	310,681	775,834,643	98.53%
Tiktok	3656	7085	1,253,112	95.51%
MovieLens	6040	3706	1,000,209	95.53%

- **Tiktok dataset**[1]: Tiktok datasets consist of a lot of micro-videos whose duration is range from 3–15 s. It also contains user attribute features (such as user id, and city), micro-video multi-modal features (such as title, and background music), and user-video historical interaction (such as click and like) for research purposes. We use the multi-modal feature vectors in original datasets without providing raw data.
- **Kwai dataset**[2]: Similar to the Tiktok dataset, Kwai is a large-scale dataset published by Kwai (a micro-video sharing and creating medium) which contains several micro-videos and users. It also contains the attribute information of users and micro-videos, and their interactive behavior (e.g., user clicks micro-videos). This dataset does not have audio information and textual information. The total dataset contains 15,727 users, 153005 micro-videos, and 215,942 interactions respectively.
- **MovieLens (MLs) dataset**[3]: This dataset provides the rating score for movies which is widely used in recommendation evaluation. We utilize the version of one million records with user attributes in our work. If a user has interaction with an item, it will be labeled as 1. Otherwise, it will be rated as 0.

5.1.2. Baselines

We compare MPGCL with several SOTA algorithms, including a feature-based method, seven GNN-based methods, and a contrastive learning method.

DeepFM (Guo et al., 2017)**:** DeepFM is a feature-based method that models high- and low-order feature representations by leveraging factorization machines and deep neural networks simultaneously. For our micro-video recommendation task, we initialize users and micro-videos by their attribute features.

GraphSage (Hamilton et al., 2017)**:** This is a widely used graph-based inductive neural network that is designed for homogeneous graph representation learning. This method sample local neighborhoods for a node and aggregates features from neighborhood nodes to efficiently generate node embeddings. The initial features of nodes are multi-modal attributes and we update the embedding of each node by feature aggregation for a fair comparison.

MMGCN (Wei et al., 2019)**:** MMGCN is a GNN-based personalized micro-video recommendation method that models users and micro-videos as a bipartite graph and encodes their interactions by a message-passing mechanism of GCN. In particular, the modal-specific user preference is first considered in this method.

LightGCN (He et al., 2020)**:** LigntGCN is a recommendation method that utilizes an efficient and simple graph convolution network where feature transformation and nonlinear function of standard GCN are removed.

UltraGCN (Mao et al., 2021)**:** UltraGCN extends LightGCN by removing explicit message-passing layers to realize efficient recommendations. It derives a constraint loss to straightforwardly approach the limit of infinite graph convolutional layers.

HAN (Wang et al., 2019)**:** HAN is designed for heterogeneous graphs which proposes to use a graph neural network with hierarchical attention to evaluate the weight of nodes and meta-paths.

[1] http://ai-lab-challenge.bytedance.com/tce/vc/
[2] https://www.kwai.com/
[3] http://grouplens.org/datasets/movielens/1m/

Table 2
Results of MPGCL and comparison methods on three datasets in terms of four evaluation metrics. (k = 10, **training_rate = 0.8**)

Datasets	Metrics	DeepFM	GraphSage	MMGCN	LightGCN	UltraGCN	HAN	HetGNN	GCL	HHFAN	Ours
Kwai	Pre	0.1255	0.2312	0.3794	0.3999	0.4019	0.3713	0.4013	0.3622	0.4233	**0.4255**
	Rec	0.2487	0.3948	0.4211	0.4075	0.4117	0.4072	0.4205	0.3749	0.4571	**0.4601**
	NDCG	0.3431	0.4401	0.4589	0.4661	0.4674	0.4641	0.4751	0.4617	0.4923	**0.5007**
	AUC	0.6557	0.7413	0.7601	0.7552	0.7581	0.7506	0.7705	0.7264	0.7944	**0.7998**
Tiktok	Pre	0.1177	0.2139	0.3533	0.3399	0.3589	0.3603	0.3741	0.2961	0.3833	**0.3905**
	Rec	0.2461	0.3751	0.3914	0.4016	0.4105	0.3636	0.3997	0.3595	0.4484	**0.4553**
	NDCG	0.3445	0.4421	0.4459	0.4512	0.4429	0.4491	0.4573	0.4029	0.4697	**0.4798**
	AUC	0.6351	0.7106	0.7379	0.7392	0.7419	0.7359	0.7422	0.7005	0.7601	**0.7686**
MLs	Pre	0.1527	0.2193	0.3312	0.3531	0.3537	0.3622	0.3604	0.3239	0.3701	**0.3822**
	Rec	0.2593	0.3546	0.3602	0.3557	0.3691	0.3693	0.3991	0.3424	0.4179	**0.4269**
	NDCG	0.3881	0.4215	0.4311	0.4323	0.4371	0.4374	0.4461	0.4019	0.4555	**0.4662**
	AUC	0.6747	0.6994	0.7037	0.7199	0.7167	0.7247	0.7383	0.6981	0.7466	**0.7628**

Table 3
Results of MPGCL and comparison methods on three datasets in terms of four evaluation metrics. (k = 10, **training_rate = 0.6**)

Datasets	Metrics	DeepFM	GraphSage	MMGCN	LightGCN	UltraGCN	HAN	HetGNN	GCL	HHFAN	Ours
Kwai	Pre	0.1812	0.3221	0.3755	0.3779	0.3815	0.3827	0.3936	0.3261	0.4047	**0.4173**
	Rec	0.2733	0.3694	0.4035	0.3827	0.3891	0.3897	0.4101	0.3718	0.4225	**0.4341**
	NDCG	0.3491	0.4113	0.4349	0.4214	0.4390	0.4392	0.4453	0.3784	0.4649	**0.4722**
	AUC	0.6910	0.6999	0.7225	0.7238	0.7241	0.7226	0.7330	0.7041	0.7499	**0.7629**
Tiktok	Pre	0.1551	0.2512	0.3001	0.2991	0.3299	0.3301	0.3302	0.2619	0.3455	**0.3611**
	Rec	0.2444	0.3367	0.3661	0.3892	0.3725	0.3891	0.3924	0.3262	0.4051	**0.4141**
	NDCG	0.3558	0.3812	0.4003	0.4303	0.4299	0.4301	0.4542	0.3991	0.4658	**0.4799**
	AUC	0.6522	0.7185	0.7291	0.7193	0.7311	0.7340	0.7344	0.7004	0.7331	**0.7427**
MLs	Pre	0.1666	0.2662	0.3117	0.3229	0.3317	0.3348	0.3367	0.3101	0.3499	**0.3627**
	Rec	0.2793	0.3351	0.3744	0.3794	0.3795	0.3703	0.3999	0.3447	0.4031	**0.4159**
	NDCG	0.3997	0.4033	0.4021	0.4104	0.4197	0.4194	0.4240	0.3792	0.4239	**0.4401**
	AUC	0.6704	0.6884	0.7001	0.7031	0.7063	0.7191	0.7014	0.6800	0.7341	**0.7501**

Table 4
Results of MPGCL and comparison methods on three datasets in terms of four evaluation metrics. (k = 10, **training_rate = 0.4**)

Datasets	Metrics	DeepFM	GraphSage	MMGCN	LightGCN	UltraGCN	HAN	HetGNN	GCL	HHFAN	Ours
Kwai	Pre	0.1922	0.2551	0.3741	0.3827	0.3841	0.3887	0.4001	0.3139	0.4142	**0.4237**
	Rec	0.2705	0.3603	0.4151	0.3999	0.3928	0.3926	0.4322	0.3722	0.4469	**0.4603**
	NDCG	0.3561	0.4229	0.4605	0.4599	0.4607	0.4566	0.4610	0.4040	0.4611	**0.4729**
	AUC	0.6701	0.7205	0.7340	0.7399	0.7421	0.7425	0.7538	0.7229	0.7641	**0.7783**
Tiktok	Pre	0.1428	0.2821	0.3553	0.3667	0.3790	0.3694	0.3711	0.3005	0.3816	**0.3949**
	Rec	0.2433	0.3804	0.3881	0.3894	0.4021	0.4006	0.4115	0.3793	0.4259	**0.4441**
	NDCG	0.3504	0.3669	0.4233	0.4305	0.4340	0.4341	0.4431	0.3883	0.4555	**0.4662**
	AUC	0.6783	0.6902	0.7336	0.7341	0.7209	0.7251	0.7332	0.7005	0.7395	**0.7547**
MLs	Pre	0.1801	0.3003	0.3294	0.3295	0.3403	0.3339	0.3505	0.3004	0.3670	**0.3799**
	Rec	0.2883	0.3297	0.3773	0.3804	0.3839	0.3827	0.3994	0.3162	0.4147	**0.4252**
	NDCG	0.3781	0.3412	0.4298	0.4304	0.4371	0.4298	0.4403	0.3884	0.4535	**0.4607**
	AUC	0.6693	0.6511	0.6990	0.6934	0.6884	0.6802	0.6994	0.7003	0.7105	**0.7339**

HetGNN (Zhang et al., 2019)**:** HetGNN aims to learn node representations on heterogeneous graphs by modeling relations of heterogeneous features. It learns node embedding by aggregating content embeddings of type-specific neighboring groups.

GraphCL (GCL) (You et al., 2020)**:** GCL is a self-supervised method for heterogeneous graph representation learning, which is designed to learn representations by maximizing consistency between different graph augmentations by contrastive learning.

HHFAN (Cai et al., 2022)**:** HHFAN is a state-of-the-art micro-video recommendation method. This method aims to generate high-quality embeddings by modeling complicated interactions of users, micro-videos, and their corresponding attributes on heterogeneous graphs.

5.1.3. Evaluation metrics and implementation details

Four evaluation metrics, including Precision@K (Pre@K), Recall@K (Rec@K), NDCG@K, and Area Under Curve (AUC) are used to evaluate our method and the baselines. K is set to 10 in our work and we demonstrate the average scores on the testing dataset.

The code of all methods is implemented by Tensorflow, a deep learning framework, and conducted on an NVIDIA GTX 3060 GPU. The parameter setting is introduced as follows:

Following experimental settings of the popular heterogeneous graph representation learning model HetGNN (Zhang et al., 2019) and the SOTA micro-video recommendation model HHFAN (Cai et al., 2022), each dataset is randomly split into the training set and the testing set. We chose p% of the data for training and the remaining (1-p)% for testing. p% is the training rate which means the percentage of training data. And p% varies from 80% to 40%. We set different train-test split ratios (0.8/0.2, 0.6/0.4, and 0.4/0.6) to simply simulate the ratios between training datasets and test datasets in different recommendation scenarios. In addition, in order to choose hyperparameters more reasonably, a validation set should be set in experiments. So we add comparison experiments that randomly split datasets into training, validation, and testing sets with a 0.8:0.1:0.1 ratio.

For the training dataset, we sample negative user-video pairs randomly. Our model is optimized by Adam optimizer. The model parameters are randomly initialized by a Gaussian distribution with a mean of 0 and a standard deviation of 0.02. We fix the embedding dimension of all models to 200 for a fair comparison. The learning rate r is selected from the set $\{10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}\}$ and the regularization weight λ

Table 5
Results of MPGCL and comparison methods on three datasets in terms of four evaluation metrics. (k = 10, train:eval:test = 0.8:0.1:0.1)

Datasets	Metrics	DeepFM	GraphSage	MMGCN	LightGCN	UltraGCN	HAN	HetGNN	GCL	HHFAN	Ours
Kwai	Pre	0.3373	0.3549	0.3703	0.3694	0.3720	0.3779	0.3809	0.3710	0.3801	**0.3990**
	Rec	0.3002	0.3559	0.4339	0.4573	0.4581	0.4696	0.4695	0.4341	0.4786	**0.5038**
	NDCG	0.4091	0.4582	0.4881	0.5031	0.5002	0.5397	0.5568	0.4785	0.5596	**0.5813**
	AUC	0.6671	0.6998	0.7118	0.7279	0.7331	0.7498	0.7530	0.7078	0.7528	**0.7793**
Tiktok	Pre	0.2514	0.2929	0.2918	0.3094	0.3200	0.3196	0.3293	0.2858	0.3295	**0.3546**
	Rec	0.3667	0.3686	0.4028	0.4381	0.4475	0.4628	0.4702	0.4114	0.4707	**0.4821**
	NDCG	0.3878	0.3871	0.3997	0.4108	0.4123	0.4112	0.4194	0.4011	0.4239	**0.4527**
	AUC	0.6493	0.6997	0.7044	0.7263	0.7314	0.7553	0.7708	0.6994	0.7704	**0.7937**
MLs	Pre	0.2323	0.3172	0.3555	0.3693	0.3782	0.3774	0.3812	0.3448	0.3803	**0.4091**
	Rec	0.3008	0.3491	0.3889	0.4198	0.4239	0.4295	0.4294	0.3994	0.4405	**0.4636**
	NDCG	0.3991	0.4100	0.4319	0.4474	0.4599	0.4691	0.4784	0.4223	0.4813	**0.5038**
	AUC	0.6883	0.7290	0.7383	0.7499	0.7681	0.7661	0.7682	0.7094	0.7790	**0.8121**

Table 6
Comparison results of various variants under different datasets. (k = 10, **training_rate = 0.8**)

Datasets	Metrics	MPGCL¬gg	MPGCL¬ms	MPGCL¬cl	MPGCL
Kwai	Pre	0.3891	0.3919	0.4011	**0.4255**
	Rec	0.4439	0.4471	0.4383	**0.4601**
	NDCG	0.4668	0.4725	0.4881	**0.5007**
	AUC	0.7609	0.7551	0.7572	**0.7998**
Tiktok	Pre	0.3737	0.3828	0.3784	**0.3905**
	Rec	0.4149	0.4473	0.4402	**0.4553**
	NDCG	0.4558	0.4635	0.4704	**0.4798**
	AUC	0.7417	0.7548	0.7592	**0.7686**
MLs	Pre	0.3558	0.3709	0.3742	**0.3822**
	Rec	0.4093	0.4092	0.4208	**0.4269**
	NDCG	0.4295	0.4523	0.4520	**0.4662**
	AUC	0.7397	0.7489	0.7504	**0.7628**

Table 7
Comparison results of various variants under different datasets. (k = 10, **training_rate = 0.6**)

Datasets	Metrics	MPGCL¬gg	MPGCL¬ms	MPGCL¬cl	MPGCL
Kwai	Pre	0.3999	0.4047	0.4029	**0.4173**
	Rec	0.4228	0.4147	0.4115	**0.4341**
	NDCG	0.4494	0.4624	0.4681	**0.4722**
	AUC	0.7483	0.7462	0.7536	**0.7629**
Tiktok	Pre	0.3451	0.3495	0.3593	**0.3611**
	Rec	0.3994	0.4036	0.4007	**0.4141**
	NDCG	0.4558	0.4652	0.4685	**0.4799**
	AUC	0.7207	0.7248	0.7381	**0.7427**
MLs	Pre	0.3394	0.3524	0.3401	**0.3627**
	Rec	0.3874	0.3926	0.4022	**0.4159**
	NDCG	0.4118	0.4304	0.4325	**0.4401**
	AUC	0.7199	0.7394	0.7336	**0.7501**

Table 8
Comparison results of various variants under different datasets. (k = 10, **training_rate = 0.4**)

Datasets	Metrics	MPGCL¬gg	MPGCL¬ms	MPGCL¬cl	MPGCL
Kwai	Pre	0.3999	0.3914	0.4034	**0.4237**
	Rec	0.4384	0.4337	0.4421	**0.4603**
	NDCG	0.4552	0.4493	0.4528	**0.4729**
	AUC	0.7381	0.7299	0.7553	**0.7783**
Tiktok	Pre	0.3562	0.3582	0.3554	**0.3949**
	Rec	0.4133	0.4002	0.4215	**0.4441**
	NDCG	0.4399	0.4329	0.4551	**0.4662**
	AUC	0.7283	0.7201	0.7228	**0.7547**
MLs	Pre	0.3499	0.3554	0.3527	**0.3799**
	Rec	0.3867	0.3894	0.4003	**0.4252**
	NDCG	0.4295	0.4407	0.4391	**0.4607**
	AUC	0.7009	0.7138	0.7114	**0.7339**

with L2 is selected from the set $\{10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}\}$. The hyper-parameters $\alpha = 0.6$, $\beta = 1$, and $\gamma = 0.5$. The max length of meta-path ml is set to 7.

5.2. Overall comparison

To evaluate the improvements of MPGCL, we compare it with the baselines mentioned above. Tables 2, 3, and 4 demonstrate the results of our method and the baselines with the training rate set by 80%, 60%, and 40%, respectively. The main comparison analyses are as follows:

Among all the baselines, DeepFM performs worse than other baselines on most cases. It is because that DeepFM is a feature-based method that only utilizes simple feature interaction extracted in user and micro-video attributes. In fact, there are complex feature interactions in micro-video recommendation systems. Therefore, this method cannot work well on micro-video recommendation.

Compared with the DeepFM method, GNN-based methods including homogeneous graph embedding based method (GraphSAGE), bipartite graph embedding-based methods (MMGCN, LightGCN, and UltraGCN), and heterogeneous graph embedding-based methods (HAN HetGNN, and HHFAN) are consistently better on all datasets. The results indicate the effectiveness of utilizing GNNs for recommendation systems, as GNNs could capture high-order interactions among users and micro-videos for high-quality embeddings. This is also the reason that we chose GNN as the base encoder for our method.

Bipartite graph embedding and heterogeneous graph embedding-based method beat GraphSAGE on three datasets. This is because GraphSAGE is designed for homogeneous graph representation learning while ignoring the heterogeneity of the user-micro-video bipartite graph. In contrast, Bipartite graph embedding and heterogeneous GNN can jointly consider the heterogeneity of graph nodes and heterogeneous contents of attributes effectively. The results also demonstrate that, among all GNN-based baselines, the recently proposed HHFAN method achieves better performance. These results indicate that leveraging heterogeneity and complex interaction both contribute to the performance of recommendations.

GraphCL is competitive with the results of the GNN-based models, even though it ignores the heterogeneity of the graph. We could conclude that GraphCL has the advantage of improving the performance of recommendation by learning robust node representations, while the other GNN-based methods ignore robustness.

We empirically find that our model outperforms all the baseline methods on three datasets with all evaluation metrics. In particular, achieve great improvements to the best baseline HHFAN, which indicates the effectiveness of our method on micro-video recommendation systems. We attribute such superiority of our method to the following reasons: (1) Compared with DeepFM, which only considers simple user-video interaction or feature interaction, we model high-order interactions among users and micro-videos by aggregating neighborhood features along a constructed bipartite graph. Therefore, we can preserve more semantic information in node representation. (2) Compared with the homogeneous graph embedding-based method (GraphSAGE)

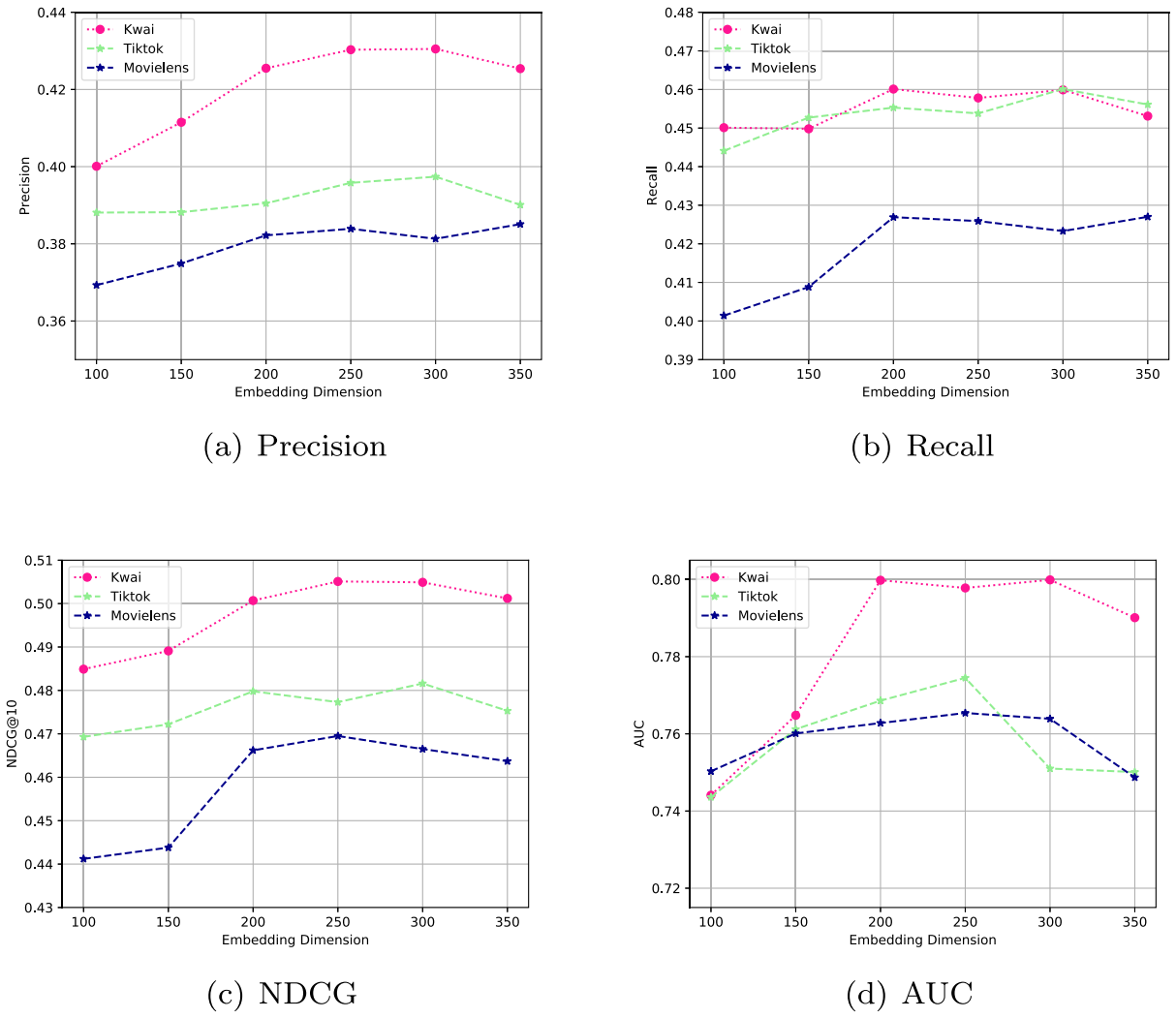


Fig. 3. Results comparison when varying parameter **embedding dimension d** of MPGCL under different metrics on three datasets. (K = 10, training_rate = 0.8).

and the contrastive learning method (GraphCL), we could find that considering the heterogeneity of user-video graphs can enhance recommendation performance. (3) Compared with all GNN-based methods, our proposed method can learn robust representations by maximizing consistency between different augmentations of graphs. We could also empirically conclude the advantage of introducing contrastive learning to the micro-video recommendation system.

We can conclude that our model can achieve consistent best results under different train–test split ratios (different recommendation scenarios), which shows that our model can effectively model heterogeneity and obtain robust representations based on contrastive learning mechanisms.

In addition, Table 5 demonstrates the results of our method and the baselines with training, validation, and testing sets divided at the ratio of 0.8:0.1:0.1. From the results, we can draw the consistent conclusion that our model achieves the best performance on all datasets. The results also indicate that our model can better model heterogeneity and obtain robust representations based on contrastive learning mechanisms.

5.3. Ablation analysis

To better understand each component of the MPGCL, we conducted ablation studies to show how each component impacts performance.

- MPGCL¬gg: This variant of our method aims to demonstrate the impact of GNN-based graph representation learning network, which is implemented by removing this network and just considering a meta-path-based Graph Contrastive Learning Network for learning representations of users and micro-videos.
- MPGCL¬ms: This variant of our method does not consider the meta-path-based random walk method. We do not generate type-specific homogeneous graphs and just leverage user-video bipartite interaction graphs for our well-designed graph contrastive learning network.
- MPGCL¬cl: This variant of our method does not consider the graph contrastive learning network. Instead, GCN networks are applied to learning each node representation of all type-specific homogeneous graphs for recommendation.

Tables 6–8 illustrate the ablation study results on all datasets in terms of four evaluation metrics. From the results, we can find that: (1) Compared with MPGCL¬gg, MPGCL has achieved better results. The reason is that, in addition to modeling heterogeneity, MPGCL also captures feature interactions of users and micro-videos from different views, thus obtaining more robust node representations. (2) The results of MPGCL outperform MPGCL¬ms which verifies the effectiveness of the meta-path-based strategy. In other words, the results indicate that modeling heterogeneity can boost the performance of recommendations. (3) The contrastive learning network aims to learn robust

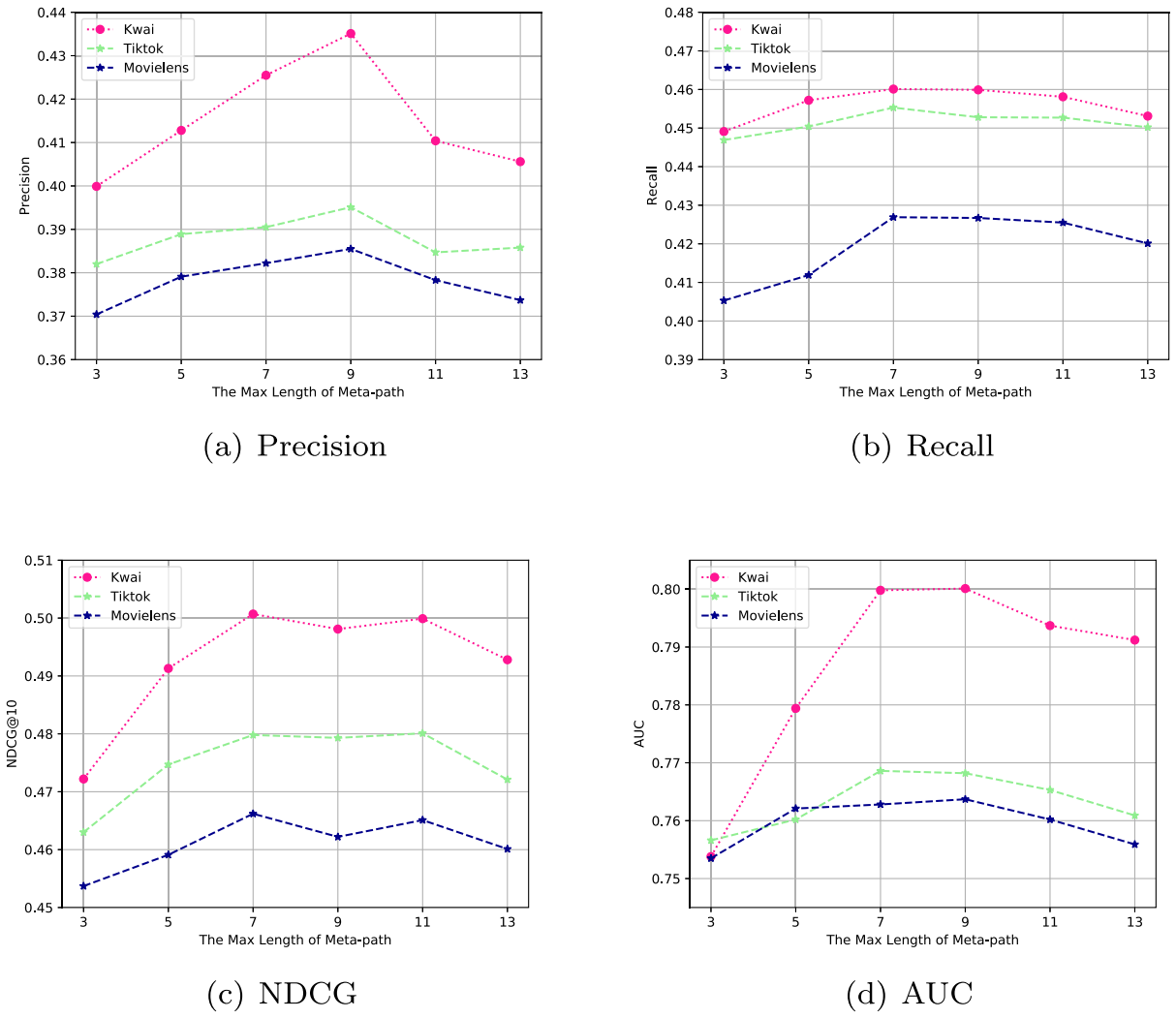


Fig. 4. Results comparison when varying parameter **the max length of meta-path** of MPGCL under different metrics on three datasets. (K = 10, training_rate = 0.8).

representations by maximizing the consistency between different graph augmentations. Our MPGCL outperforms the MPGCL$\neg cl$, from which we can conclude that introducing contrastive learning networks can help to obtain robust node representations and boost recommendation.

From the above discussion, we can draw the conclusion that our method achieves great benefits with the incorporation of three components.

5.4. Parameter sensitivity

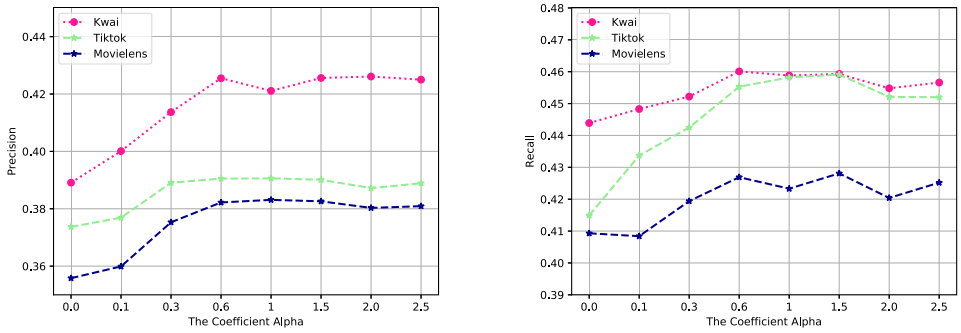
We study the impact of different hyperparameters on micro-video recommendation including dimension d, α, β, γ, the ranking position K, and the max length of meta-path. The extensive experiments are done on Kwai, Tiktok, and MovieLens datasets and report the results under four metrics. The training rate is fixed at 0.8. We demonstrate the results in Figs. 3–8.

To verify the impact of dimensions on the results, we first vary the embedding dimensions d of nodes in the set of $\{100, 150, 200, 250, 300, 350\}$ to show the influences. And Fig. 3 shows the results of our method under different dimensions. We can see that the performance of our method consistently increases when d changes from 100 to 200. However, the performance slightly decreases when we continuously increase the number of dimensions. The reason may be that large dimension carry redundant information, which may cause overfitting and decrease the performance of recommendation systems.

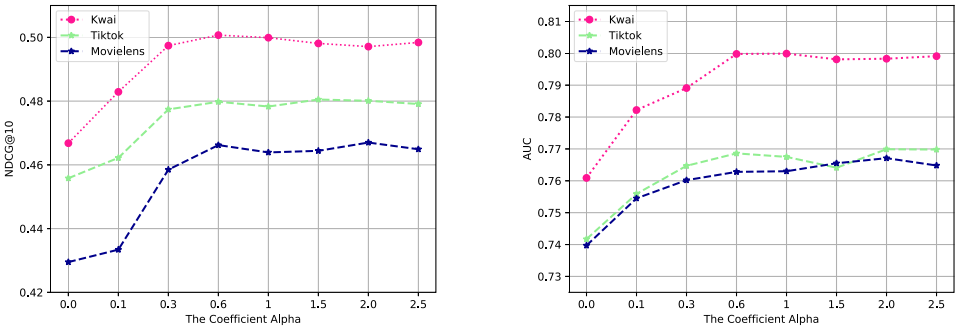
Further, we conduct experiments with different max lengths of meta-path to study its influence. The results are demonstrated in Fig. 4. We can observe that the setting ml = 7 has achieved the best performance in most cases. This suggests that with the increase of ml, our model can learn more high-order information. When ml is large enough, the model is not performing well. We assume that our model may suffer from over-smoothing.

Next, we investigate the impact of α, β, and γ. First, we fix $\beta = 0$, $\gamma = 0$ and tune the weight α of the GNN loss. The number of α is varied in the range of $\{0.0, 0.1, 0.3, 0.6, 1.0, 1, 5, 2.0, 2, 5\}$. As shown in 5, the best performance of our model is obtained at 0.6. Then, we fix $\alpha = 0$, $\gamma = 0$ and tune the weight β to show the impact of the CL part. As shown in Fig. 6, when β changes from 0.1 to 2.5, the performance of our model increases steadily at first and then decreases, which indicates that the CL part is valuable for recommendation. It can be found that $\beta = 1$ is the best choice for our model. We also vary γ in the range of $\{0.1, 0.3, 0.5, 0.7, 0.9, 1.1\}$ to evaluate its effectiveness. As shown in Fig. 7, the best performance is obtained when γ is set to 0.5. The results show that the proposed three parts are important for improving the performance of micro-video recommendations.

Finally, we conduct experiments to analyze the impacts of the ranking position K of our proposed model MPGCL and all baselines. Fig. 8 shows the experimental results of Top-K recommendations in terms of NDCG@K and AUC@K on three datasets respectively, where the ranking position K ranges from 2 to 14. We can conclude that our

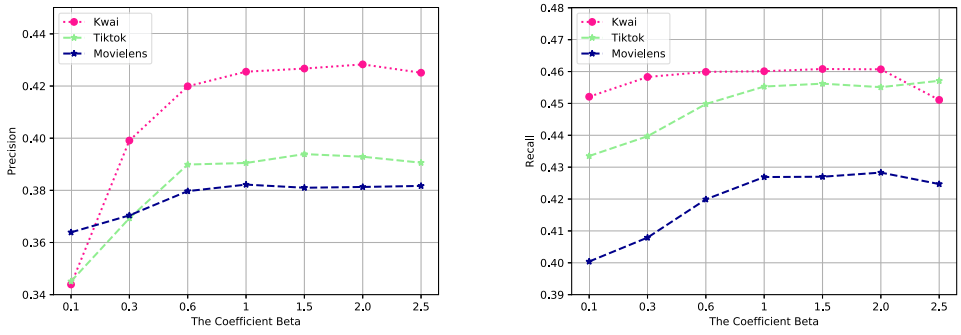


(a) Precision (b) Recall

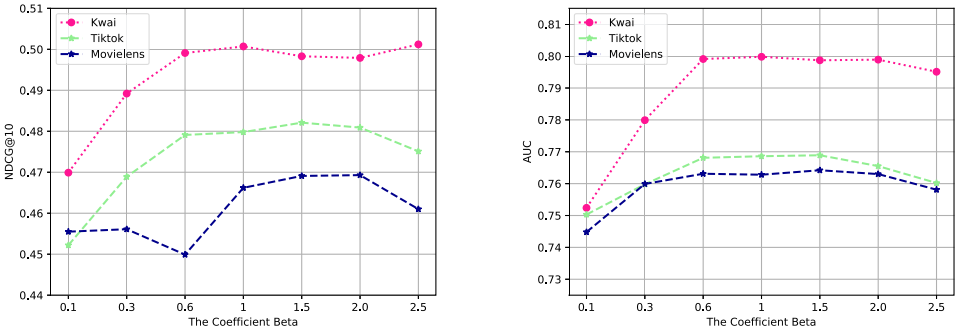


(c) NDCG (d) AUC

Fig. 5. Results comparison when varying parameter α of MPGCL under different metrics on three datasets. K = 10, training_rate = 0.8.



(a) Precision (b) Recall



(c) NDCG (d) AUC

Fig. 6. Results comparison when varying parameter β of MPGCL under different metrics on three datasets. K = 10, training_rate = 0.8.

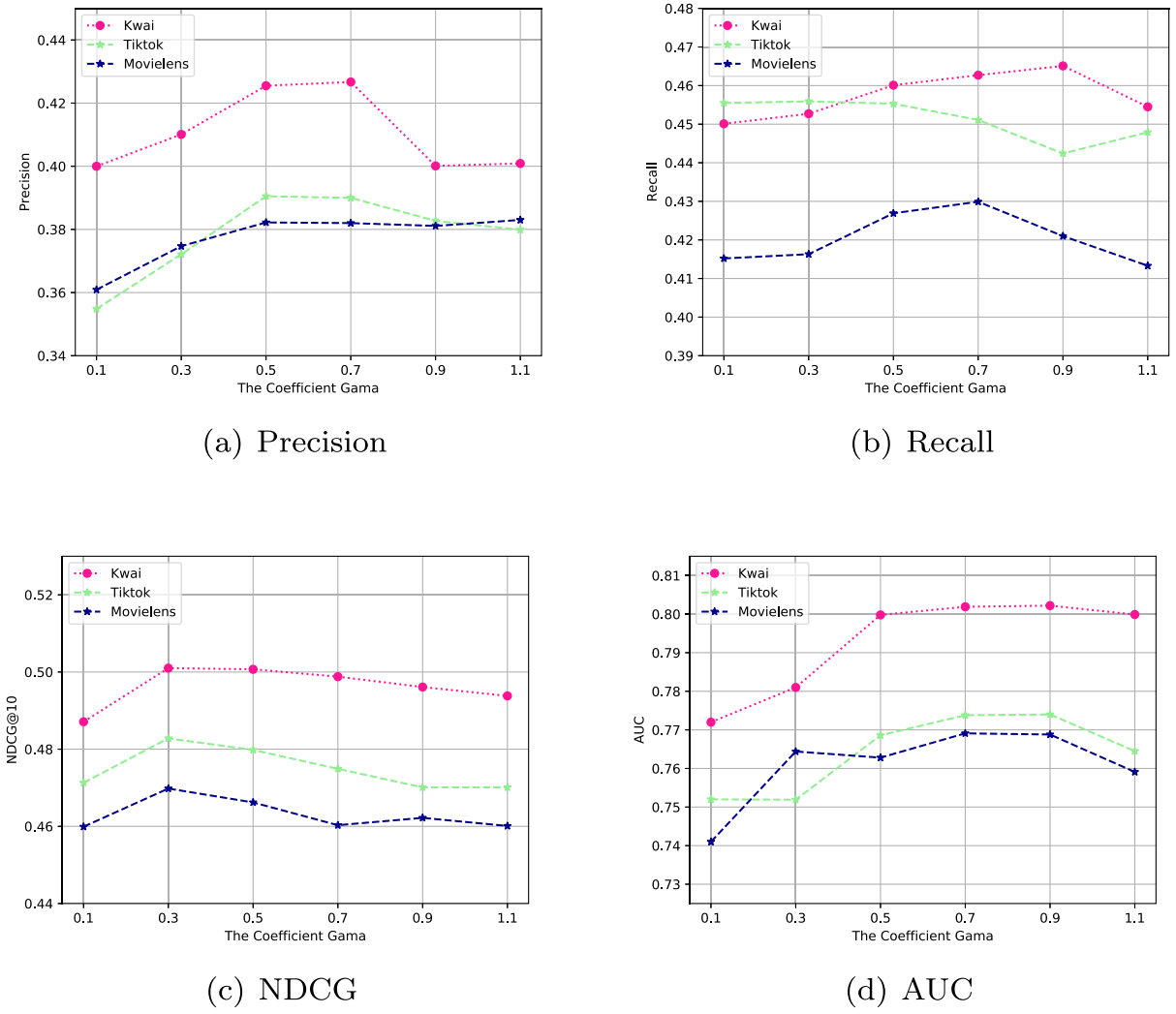


Fig. 7. Results comparison when varying parameter γ of MPGCL under different metrics on three datasets. K = 10, training_rate = 0.8.

proposed MPGCL has consistent improvements over other baselines and more stable performances than other baselines across positions, indicating its better capacity for modeling the heterogeneity of heterogeneous graphs and learning more expressive embeddings based on contrastive learning mechanisms.

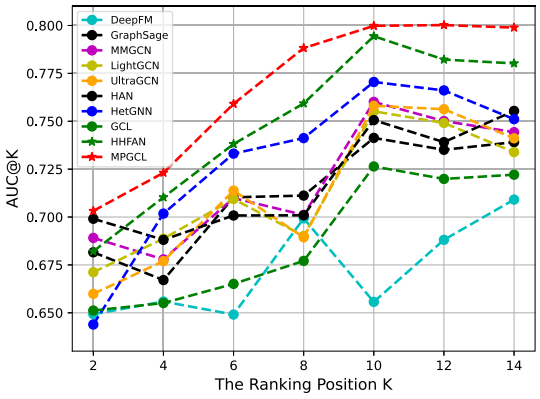
5.5. Statistical test

In order to verify the significance of the model, in terms of NDCG and AUC metrics, we perform t-test (Student's t test) between our proposed MPGCL and each baseline (DeepFM, GraphSage, MMGCN, LightGCN, UltraGCN, HAN, HetGNN, GCL, HHFAN), respectively, on three datasets. Note that the training rate is fixed as 0.8, and the ranking position K is set to 10. And the heatmap result of t-test is shown in Fig. 9: The abscissa of the heatmap is the baselines, and the ordinate represents the combination of three datasets and two metrics (NDCG and AUC). For example, a row of data corresponding to Kwai-NDCG in the heatmap indicates the t-test results of MPGCL and each baseline, on the Kwai dataset in terms of NDCG. From Fig. 9, we can see that all t-test results are less than 0.05, demonstrating that our proposed model significantly outperforms baselines.

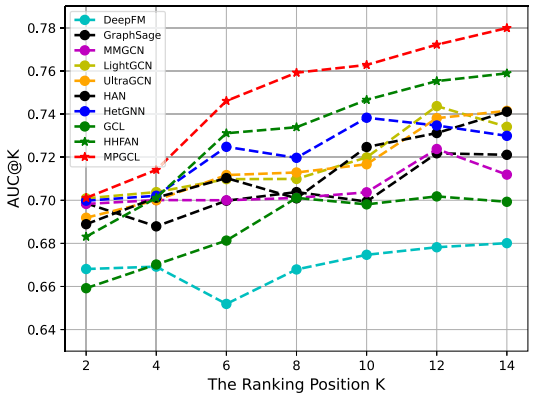
5.6. Cold-start recommendation

Our proposed model still has certain applicability for cold-start recommendation tasks including user cold-start recommendation tasks and micro-video cold-start recommendation tasks. According to whether there are attributes and whether there are interactions, cold-start recommendation tasks can be divided into four categories: (1) There are attributes and a small number of interactions (The number of interactions is not more than five): We can construct user-video bipartite graphs with node embeddings, and directly use our well-trained model to infer representations of cold-start users or cold-start micro-videos, and further use these representations for cold-start recommendation tasks. (2) There are no attributes and only a small number of interactions (The number of interactions is not more than five): We can construct user-video bipartite graphs without node embeddings (zero vectors), and further utilize our well-trained model to infer representations of cold-start users or cold-start micro-videos for cold-start recommendations. (3) There are only attributes and no interactions: We cannot build user-video bipartite graphs to use the trained model so we can only infer representations of cold-start users or cold-start micro-videos through their attributes. (4) There are no attributes and no interactions: In such cases, we cannot do cold-start recommendation tasks.

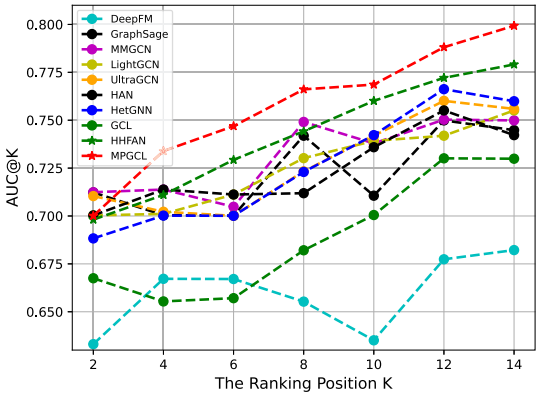
To verify the effectiveness of our model on cold-start recommendation tasks, compared to more powerful graph-based baselines, we conduct user cold-start experiments in the first type of cold-start recommendation category (There are attributes and a small number of interactions). The results in terms of NDCG and AUC metrics on three datasets are shown in Table 9.



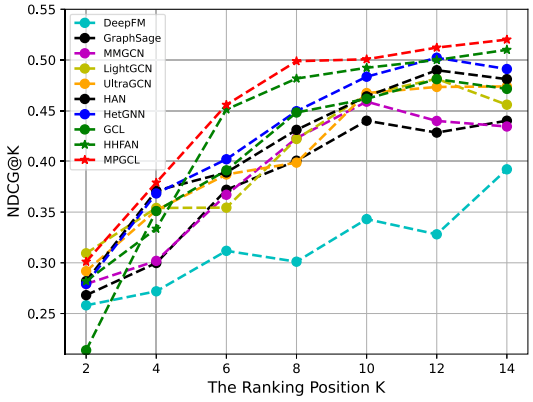
(a) AUC on Kwai dataset



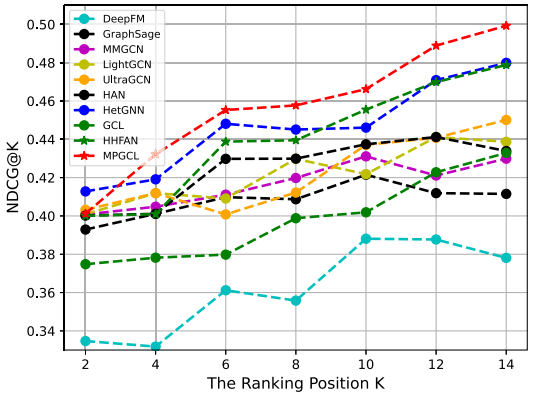
(b) AUC on MLs dataset



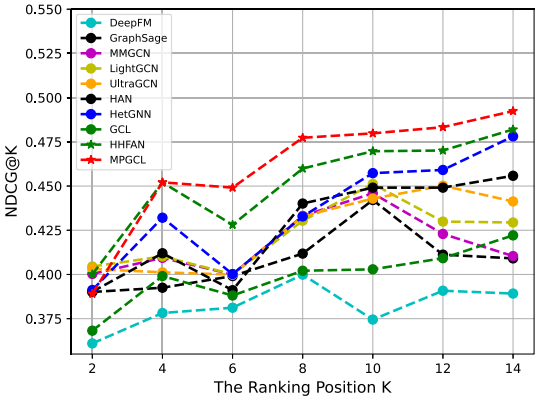
(c) AUC on Tiktok dataset



(d) NDCG on Kwai dataset



(e) NDCG on MLs dataset



(f) NDCG on Tiktok dataset

Fig. 8. Results comparison when varying parameter **K** of MPGCL under different metrics on three datasets. training_rate = 0.8.

From the experimental results in Table 9, we can see that our proposed model has achieved consistent superiority over other graph-based baselines in the cold-start recommendation tasks.

6. Conclusions

In our work, we propose a micro-video recommendation method, named meta-path-based graph contrastive learning network (MPGCL), which contains a meta-path-based random walk method and a graph contrastive learning network. We do extensive experiments on three datasets, and the results demonstrate that our model outperforms the baselines. In the future, our work will consider two aspects for improvement. First, we will take more multi-modal features into consideration and consider how to better model their interactions. Second, we will

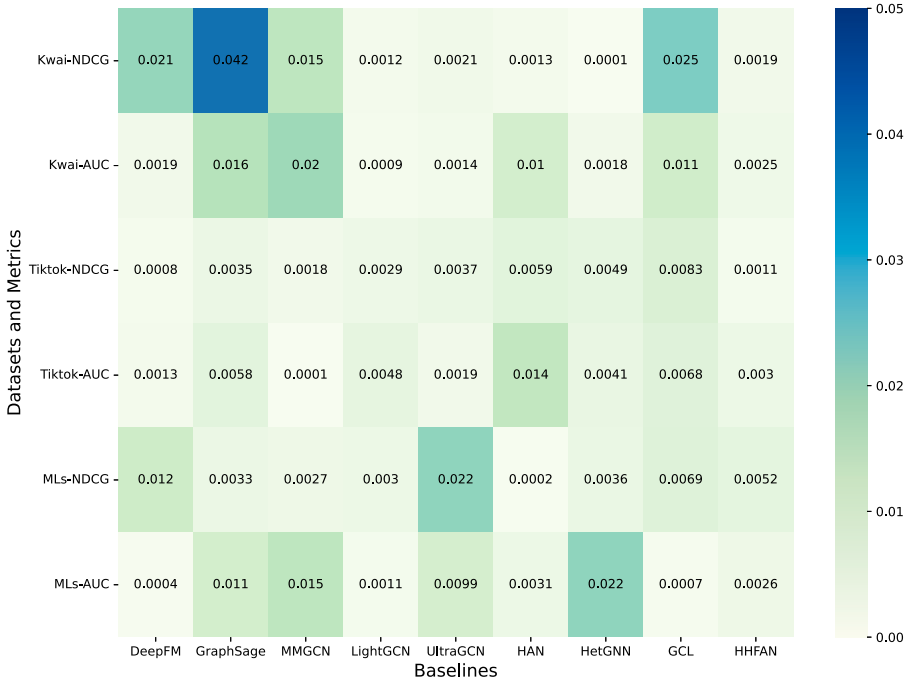


Fig. 9. Heatmap result of t-test.

Table 9
Results of MPGCL and comparison methods on three datasets in terms of two evaluation metrics for **user cold-start recommendation.** (k = 10, training_rate = 0.8)

Datasets	Metrics	MMGCN	LightGCN	UltraGCN	HAN	HetGNN	HHFAN	Ours
Kwai	NDCG	0.2999	0.3111	0.3011	0.3384	0.3449	0.3599	**0.3822**
	AUC	0.6649	0.6555	0.6558	0.6647	0.6738	0.6869	**0.7004**
Tiktok	NDCG	0.2893	0.2889	0.3132	0.3009	0.3195	0.3348	**0.3595**
	AUC	0.6789	0.6946	0.6899	0.7031	0.7231	0.7201	**0.7338**
MLs	NDCG	0.2458	0.2339	0.2510	0.2669	0.2788	0.2789	**0.3001**
	AUC	0.6786	0.7009	0.7142	0.7115	0.7284	0.7298	**0.7537**

consider more relational features between users and micro-videos, such as the inter-relation between them.

CRediT authorship contribution statement

Ying He: Conceptualization, Methodology, Writing – original draft. **Gongqing Wu:** Conceptualization, Writing – original draft, Resources, Project administration, Funding acquisition. **Desheng Cai:** Software, Visualization. **Xuegang Hu:** Supervision, Writing – reviewing and editing, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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